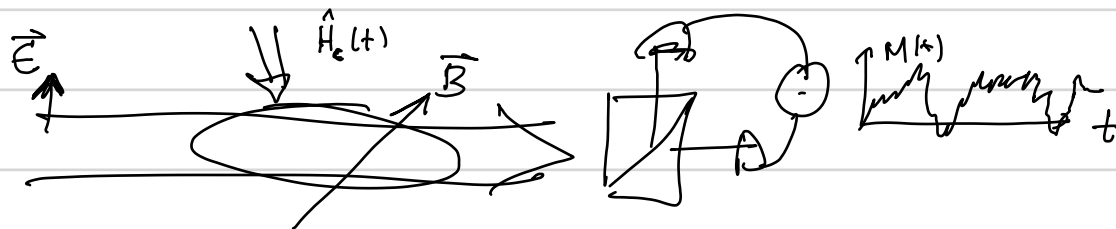


Vector Magnetometry

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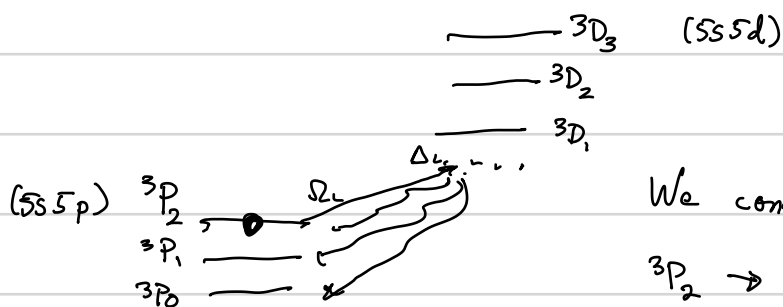
We are proposing a model for vector magnetometry based on continuous control + measurement



We want to measure an unknown $\vec{B}$ by the measuring the Larmor precession of atomic magnetic dipoles continuously through the Faraday effect. In addition to the Larmor precession the spins are subjected to dynamics induced by the light shift induced by the probe laser as well as an addition control Larmor precession. The idea is that the induced dynamics on the magnetic dipole allows us to most efficiently sense $\vec{B}$ by moving it ergodically over the sphere.

The total Hamiltonian acting on the atomic spins $\hat{H} = \hat{H}_c(t) - \gamma \vec{J} \cdot \vec{B}$
 where $\hat{H}_c(t) = \Omega (\cos\phi(t) \hat{J}_x + \sin\phi(t) \hat{J}_y) + \hat{H}_{LS}$ is the control Hamiltonian
 and $\hat{H}_{LS} = V |\vec{E} \cdot \vec{J}|^2$, for linear polarization, and $V = C^{(2)} V_0$ is tensor light shift. $C^{(2)}$ is the "tensor coefficient" from the lang paper and $V_0 = \frac{\Omega_c^2}{4\Delta_c}$ is the light shift for the probe laser with Rabi frequency Ω_c , and detuning Δ_c .

For concreteness, consider ^{88}Sr atoms prepared in the metastable $(5s5p) ^3P_2$ state. These atoms have spins $J=2$



We consider the laser probe detuned from the $^3P_2 \rightarrow ^3D_1$ transition. The 3D_1 decays $\sim 99\%$ to 3P_1 or 3P_0 and only $\sim 1\%$ to 3P_2 . Thus the effect of decoherence is mostly atom loss

$(S^2) \frac{1}{2}$ —

The time evolution of the system, thus is governed by the nonHermitian "effective Hamiltonian"

$$\hat{H}_{\text{eff}} = \hat{H}_L(t) - \vec{\omega} \cdot \hat{\vec{J}} - i \frac{\gamma_s}{2} (C^{(0)} \hat{1} + C^{(2)} |\vec{\epsilon} \cdot \hat{\vec{J}}|^2)$$

where $\gamma_s = \frac{\Omega_L^2}{4\Delta^2} \Gamma = V_0 \left(\frac{\Gamma}{\Delta} \right)$ is the photon scattering rate, $\Gamma = \text{rate of decay } ^3D_1$,

$$\Rightarrow \hat{H}_{\text{eff}} = -\vec{\omega} \cdot \hat{\vec{J}} + \Omega (\cos\phi(t) \hat{J}_x + \sin\phi(t) \hat{J}_y) + C^{(2)} V_0 |\vec{\epsilon} \cdot \hat{\vec{J}}|^2 - i \frac{\gamma_s}{2} (C^{(0)} \hat{1} + C^{(2)} |\vec{\epsilon} \cdot \hat{\vec{J}}|^2)$$

For $\omega \quad J=2 \rightarrow J'=1$ Transition $C^{(0)} = \frac{1}{5}$, $C^{(2)} = -\frac{1}{10}$

The details of the numbers are unimportant. The main take away is this. For a single color linearly polarized probe, there will be a nonlinear "one-axis twisting" Hamiltonian with a strength proportional to the light shift $V \sim \frac{\Omega_L^2}{\Delta_L}$. Decoherence is due to photon scattering of the probe laser. For the $^3P_2 \rightarrow ^3D_1$ transition photon scattering is 99% to other levels, and so can be considered as atom loss. In general, the loss rate is state dependent, because different m_J states scatter differently. But this is a detail we can ignore if we don't want to get in the weeds of the model. Loosely, the rates of loss = photon scattering rate $\gamma_s \sim \frac{\Omega_L^2}{\Delta^2} \Gamma$. So with $V = \frac{\beta}{2J} = \frac{\Omega_L^2}{4\Delta_L}$, $\gamma = \frac{\Omega_L^2}{4\Delta_L^2} \Gamma$, $\frac{\gamma}{V} = \frac{\Gamma}{\Delta_L}$.

That is, the rate of decoherence is $\frac{\Gamma}{\Delta_L}$ times the rate the nonlinearity for a given laser power. It seems, thus, that we can always avoid decoherence by going sufficiently off resonance, if we have enough laser power to generate the desired light shift.

However, there is a catch. The rate of measurement through the Faraday effect also depends on the rate of photon scattering. Physically, the measurement arises from the scattering of photons into the forward mode; decoherence is due to diffuse scattering in 4π . The measurement rate $R = C^{(1)} \gamma_s \frac{\sigma_{\text{scatt}}}{A}$ is rate that photons are scattered into the forward mode. It is proportional to the 4π -scattering rate γ_s , and the geometric factor $\frac{\sigma_{\text{scatt}}}{A}$, where σ_{scatt} = scattering cross section, A = beam mode area, $C^{(1)}$ = Vector light shift coefficient

All of this comes together as follows. The measurement record / atom

$$M(t_i) = e^{-\gamma t_i} N \langle \hat{J}_z(t_i) \rangle + \sigma N_i \leftarrow \text{Gaussian, mean 0, unit var}$$

↑
Noise strength = $\frac{1}{\sqrt{N \delta t}}$: Shot noise is finite δt

Or, per atom $m(t_i) = \frac{M(t_i)}{N} = \langle \hat{J}_z(t_i) \rangle + \frac{1}{\sqrt{N} \sqrt{OD} \sqrt{\delta t}} N_i$

Where $OD = \frac{N}{A} \sigma_{\text{scat}} = \text{optical depth}$

Backaction is negligible if $OD \gamma T \ll 1$ for total time T

Typically $OD \sim 1$ without heroics. Thus, if $\gamma T \gg 1$, we will decay the signal substantially, and we typically want to work in a regime $\gamma T \ll 1$

Bottom line: For a single color probe, the rate of measurement, the decoherence rate, and the tensor light shift are all related. The relationship between them depends on the intensity and detuning of the laser. Here is how we might think about these things

- The rate of decoherence = scattering rate γ_s , rate of measurement $R \sim \frac{\sigma_{\text{scat}}}{A} \gamma_s$
tensor light shift $\frac{\beta}{2J} \sim \gamma_s \left(\frac{A}{F} \right)$
- The detuning should be chosen to maximize the Fisher information. In the absence of decoherence, the precision of the measurement is limited solely by shot noise $\Delta J_z^2|_{\text{SN}} \sim \frac{1}{R T}$, where T is the total time of the measurement. However, there is a competition with decoherence. As time progresses, we lose signal exponential $M(t) \sim e^{-\gamma t}$. Thus, for a given photon scattering rate γ_s , there is an optimal time T to get the best precision, in the absence of nonlinearity. Probably $\gamma T \sim 1$. This we imply going closer to resonance to increase γ to speed up everything. However that comes at a cost, because we will heat the atoms as exert radiation pressure, which is to be avoided.
- Now, the addition of tensor light shift means we can get the info faster by going farther from resonance as β increases relative to γ_s as Δ_L increases.

